

ASHVIN SHIYANI

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SUMMARY

Computer Science student at Acadia University with hands-on experience shipping full-stack applications, Python tools, and distributed backend systems. Built a production-deployed rate limiting microservice with Redis, PostgreSQL, and JWT auth, a real-time multi-user chat system, and a desktop finance analytics app processing 10,000+ row CSV datasets. Comfortable across the stack — Node.js, React, WebSocket architecture, and Redis. Seeking a software development internship to contribute real engineering work from day one.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, HTML, CSS

Frameworks & Tools: Node.js, Express, React, Socket.io, Redis, PostgreSQL, Pandas, Matplotlib, CustomTkinter, Git, GitHub, VS Code, Linux

Concepts: OOP, Data Structures & Algorithms, REST APIs, WebSockets, Rate Limiting, JWT Authentication, Distributed Systems, SQL, Event-Driven Architecture

PROJECTS

Distributed Rate Limiter Microservice

2026

Node.js, Express, Redis, PostgreSQL, React, JWT, Lua [Github-Repo](#) || [Live-Demo](#)

- Architected a production-deployed rate limiting microservice implementing three algorithms — Fixed Window, Sliding Window, and Token Bucket — each enforced via Redis with per-key request counters and expiry logic.
- Engineered atomic Token Bucket operations using a Lua script executed inside Redis to eliminate race conditions in concurrent request handling across distributed clients.
- Built a full JWT authentication system with bcrypt password hashing, PostgreSQL user/API key storage, and a tiered access model (Free/Silver/Gold) controlling per-minute request limits.
- Shipped a React dashboard for API key management with a dark terminal UI — deployed as a static site on Render with a live backend API.
- Worked in a collaborative development environment, integrating contributions through GitHub pull requests, resolving merge conflicts, and maintaining a shared codebase throughout development and deployment.

Finance Dashboard

2026

Python, SQLite, Pandas, Matplotlib, CustomTkinter [Github-Repo](#)

- Engineered a desktop analytics tool that ingests CSV datasets with 10,000+ rows, parses transaction records with Pandas, and auto-generates financial charts via Matplotlib within 2–3 seconds of upload.
- Designed a persistent SQLite storage layer so users can revisit previously uploaded datasets without re-importing, reducing repeat processing time to near zero.
- Built the full UI using CustomTkinter, implementing a clean dark-themed dashboard with responsive chart panels and category-level expense breakdowns.

Real-Time Chat Application

2025

Node.js, Socket.io, JavaScript, HTML/CSS · [Github-Repo](#)

- Implemented event-driven WebSocket communication using Socket.io to support concurrent multi-user messaging — messages broadcast to all connected clients in under 100ms with no page refresh.

EXPERIENCE

Research Assistant

Oct 2024 – Present

Acadia University · Wolfville, NS

- Built Python scripts to automate data collection and processing pipelines, reducing manual research time significantly.
- Designed and queried structured datasets from 160+ sources, applying database thinking used in backend development.

Community Relations Representative

Jan 2026 – Present

Acadia Students' Union · Wolfville, NS

- Elected by peers to represent student interests in ASU council meetings, advocating for policy changes affecting 4,000+ students.
- Facilitate dialogue between Acadia students and Town of Wolfville officials, bridging student needs with community decision-making processes.

Campus Tour Ambassador

Sep 2025 – Present

Acadia University · Wolfville, NS

- Deliver guided tours to prospective students and families (up to 15 per session), communicating academic programs and campus culture to visitors from 10+ countries per semester.
- Represent the university's commitment to diversity and inclusion, adapting communication style for audiences from different cultural and academic backgrounds.

Shift Supervisor (Part-Time)

Promoted Jan 2026

Tim Hortons · Wolfville, NS

- Promoted from team member to shift supervisor, overseeing daily operations, cash handling, inventory checks, and training new team members.

EDUCATION

Bachelor of Computer Science

Sep 2023 – Present

Acadia University · Wolfville, NS

Dean's List 2024 & 2025